MATH2030J - Discrete Mathematics

Álvaro Mateos González

University of Michigan - Shanghai Jiao Tong University Joint Institute

alvaro.mateos@sjtu.edu.cn

Lecture 1

Outline

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Teaching Team

- Professor: Mateos González, Álvaro. alvaro.mateos@sjtu.edu.cn
  (Note: Mateos is a family name, not a middle name.)
  Lectures: Room DSY315, Mon, Wed, Thu, 08:00 - 09:40, + Weeks 1–3: Wed 14:00 - 15:40
  Office hours: 10:15-11:15 in Room 442, Longbin Building (as well as after class and by appointment).
- TA: TBD, TBD, TBD_TBD@sjtu.edu.cn
  Office hours: Location and time TBD.
- TA: TBD, TBD, TBD@sjtu.edu.cn
  Office hours: Location and time TBD.

Class Attendance Policy

- **Guiding principles:** Attend, participate and be considerate to your instructors and classmates.

  If personal circumstances compel you to contravene any of the more specific rules below, let me know well in advance and we will make sure that you can follow the course to the best of your possibilities without disturbing anybody else and without having to fear any sanctions.

- **Attendance to lectures** is mandatory, unless justified or discussed.

  You can miss 3 lectures, no questions asked. Beyond that, each two absences will lead to one grade step reduction (e.g. from B+ to B for 4 or 5 absences, or from B+ to C+ for 8 or 9 absences, *etc.*). If you must or intend to miss several lectures in a row and do not deserve to be penalised, you need my approval.

- **Attendance to recitations** is highly encouraged, but not mandatory.

  If you have constraints that prevent you from attending, please let me know. You can succeed while not participating actively in lectures and recitations, but that requires exceptional discipline and organisation.

- **Class interruptions:** Measures may be implemented throughout the semester if class is interrupted too frequently due to students arriving late, leaving early or other reasons. Those measures may include refusing that students enter the classroom too late or grade sanctions.
- **Electronic devices:** You may take notes on an electronic device. However, you may not use an electronic device to communicate during class or record your classmates or the lecture. If you feel the need to take pictures or record, raise your hand: there is usually another solution. Failure to respect those rules may result in expulsion from class or grade penalties depending on the frequency and severity of the disturbances.

The following rules have been laid down by the Academic Office:

- A student who has been absent from studies for more than one week because of illness or other emergency should consult the program advisor.
- Absence for illness should be supported by a hospital/doctor's certificate. A note that a student visited a medical facility is not sufficient excuse for missing an assignment or an exam. The note must specifically indicate that the student was incapable of completing an assignment or taking the exam due to medical problems.
- Late medical excuses must satisfy the following criteria to be valid:
    1. The problem must be confirmed by the doctor to be so severe that the student could not participate in the exam.
    2. The problem must have occurred so suddenly that it was impractical to contact me in advance.
    3. The student must be in contact with me immediately after the exam with the required documentation

Assignments and Evaluation

Homework:	4 assignments - see Canvas.
Discussion sessions:	3 assignments - see Canvas.
Midterm:	Tentatively on week 6, covering all topics until the end of week 5.
Final exam:	Held during exams week, covering all topics in the course with a higher focus on the topics not covered in the Midterm.
Final grade:	Completing the course evaluation: +2% bonus, Completing the Peer Evaluation for the coursework: +2% bonus, Participation grade: 10%, Midterm 40%, Final 50%.

Participation: 4% for contributing equally to homework assignments, 2% for lecture attendance and participation, 4% for contributing to discussion sessions.

Coursework

- Roughly one homework assignment every 2 weeks, avoiding exam weeks.
- You will be randomly assigned into homework groups of 3 students. You are expected to collaborate within your group and hand in a single, common solution.
- Each student must achieve 60% of the total coursework points by the end of the term in order to obtain a passing grade for the course. Failure to reach that threshold will result in failure of the course! However, passing that threshold, the homework grade will not be counted: it will have no effect on your course grade.
- Each member of an assignment group will receive the same number of points for each submission. In cases where one or more group members consistently do not contribute a commensurate share of the work, a TA will investigate the situation and individual group members may lose some or all of their marks.
- Do not collaborate on homework assignments with students outside of your group.

Coursework

- Late submissions will not be accepted - save for rare, well-justified exceptions.
- A penalisation of up to 10% of the awarded marks may be applied for poor presentation if your work is hard to read.
- You are encouraged to compose your coursework solutions in LaTeX. While this is optional, a 10% bonus to the awarded marks will be applied for assignments typed in LaTeX.
  LaTeXis open-source software for mathematical typesetting, and there are various implementations available. I suggest that you use Baidu or Google to find a suitable implementation for your computer and OS. LaTeXis widely used for writing theses and scientific papers, so it may be quite useful for you to learn it. You may consult the links on Canvas to get started.

Discussion sessions

- One out of every 6–7 lectures will be a discussion session dedicated to collective problem-solving, questions, *etc.* You will receive an exercise sheet the previous week indicating some warm-up exercises from the recommended literature that you should be able to tackle with ease, and a list of exercises.
- Each coursework group will choose or be assigned an exericse from the list. Each group must submit before the discussion session a .tex document, compilable using standard pdfLaTeX, stating the exercise they have been assigned, and their proposed solution. That solution must be extremely well written: this is an exercise to perfect your proofwriting.
- Make sure you don't write unnecessary statements, that the organisation of your proof is clear, that you define evey mathematical object you use, that you check the hypotheses of every theorem you use, etc.
- Your solutions will be shared with the whole class, and you may be asked to present them in front of the whole class.
- During the discussion session itself, you will be expected to collaborate in small groups and tackle the other exercises on the sheet.

Final grade

The top 6-12% of students will usually receive an A+. Below that, grades descend by mostly fixed thresholds. Other than that normalisation, your grade only depends on you! If everybody understands the concepts well and performs admirably in the exams, I will be delighted to give everyone an A.

In your learning process, use all the resources we provide: lectures, recitations, office hours, interactions with other students in person and online, recommended literature, etc.

Outline

Logistics

How to make the most of this course

Syllabus

Ch. 0. Introduction

Ch. 1. Logic and Proofs

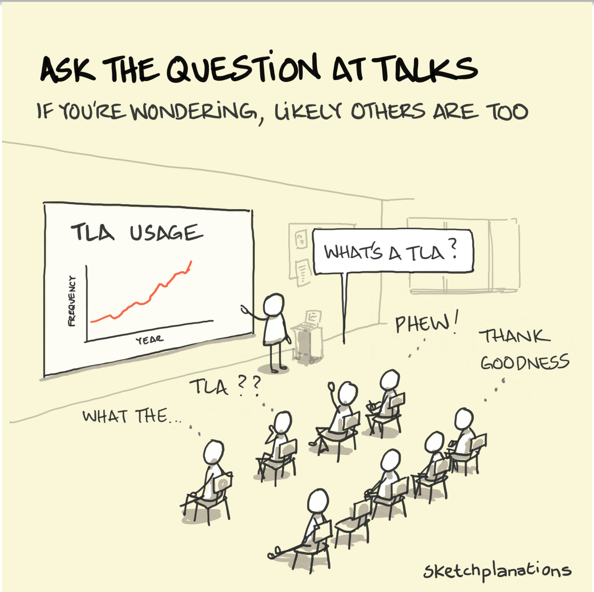
Take initiative and be proactive

- **Take notes actively:** identify the structure of the information you're learning while you're following a lecture. Your notes don't need to be exhaustive: I will share our slides. But do keep pencil and paper (or equivalent) with you during class. Examples treated on the blackboard might not be uploaded.
- **Practice active recall**: after class or recitation, close your notes and write down what you remember. Focus on definitions and general structures rather than specific details.
- **Invent your own examples and counter-examples** to fit definitions, properties, theorems. The simpler and the more general, the better! If a hypothesis of a theorem is not satisfied, why does the conclusion fail? (or why doesn't it?)
- **Review before recitations**, and make sure you know how to solve on your own some solved examples from the recommended literature.
- **Come to office hours**. Set up an appointment if you can't make it to those we set.

Ask questions often!

- Ask your own version of a question.
- Provide a big picture/background of the question.
- Provide details of your question/confusion.

Practice writing skills whenever possible: homework, emails, Piazza...



Honor code of academic integrity

- Use outside sources correctly: either understand the content and use that understanding to formulate your own answer that differs significantly from the source, or quote the source.
- AI-generated text is not a valid source.
- Tertiary sources like Wikipedia are extremely useful, but prioritise quoting secondary sources (textbooks, research articles).
- Plagiarism of sources and of other students' work will lead to heavy sanctions.
- Sharing answers during an exam, accessing and facilitating access to homework solutions to non-members of the assignment group are breaches of the honor code that can lead to heavy sanctions.

Outline

Logistics

How to make the most of this course

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Course Topics

0. **Introduction** **(1 lecture)**
   Course logistics, example problems.

1. **Logic** **(~ 5)**
   Propositional logic, truth tables, predicates, quantifiers, rules of inference, proofs.

2. **Sets** **(~ 3)**
   Definitions, set operations, functions, cardinality.

3. **Introduction to Algorithms and Induction** **(~ 5)**
   Definitions, Induction, strong induction, well-ordering, structural induction, recursive algorithms, program correctness, asymptotic notation, complexity, examples.

4. **Counting** **(~ 3)**
   Basics, Dirichlet's pigeonhole principle, permutations and combinations, inclusion-exclusion principle.

5. **Basic Number theory and Group Theory** **(~ 4)**
   Divisibility and modular arithmetic, finite magmas, monoids, groups and applications, integer representations and algorithms, primes, congruences, cryptography.

6. **Graphs and Trees** **(~ 4)**
   Basic graph theory, representation, partial order, isomorphism, connectivity, Euler and Hamilton paths, partial order and graphs, trees, tree traversal, spanning trees, Kruskal's algorithm, Dijkstra's algorithm.

Tentative Schedule

Week 1: Introduction and Logic

On Wednesday of Week 1 (Wed1) you will receive an exercise sheet for the Discussion session the following Wednesday. Each coursework group will be responsible for preparing the solution to one exercise on the sheet, writing it cleanly on LaTeX before the discussion session, and potentially presenting it in front of the class.

Due: nothing.
Handed out: DS1 (discussion sheet 1) on Wed1.
RC: no RC.

Week 2: Logic, Sets

On Wednesday of Week 2 (Wed2) you will receive HW worksheet 1, due the following Tue3 before 23:59.

Due:
- type your exercise from DS1 by Tue2,
- prepare to present during the Discussion Session on Wed2.

Handed out: HW1 on Wed2
RC: time and location TBD

Week 3: Sets, Introduction to Algorithms

Due: • HW1 on Tue3, 23:59
Handed out: DS2 on Wed3
RC: time and location TBD

Week 4: Introduction to Algorithms, Induction and Recursion

Due:
- type your exercise from DS2 by Wed4,
- prepare to present during the Discussion session on Thu4.

Handed out: HW2 on Wed4
RC: time and location TBD

Week 5: Induction and Recursion, Counting

Due: • HW2 on Thu5, 23:59
Handed out: Mock midterm = HW3 on Wed5
RC: time and location TBD

Week 6: Counting, Arithmetic, Groups, and Midterm

Mon6: half the lecture will be dedicated to student questions
Due: Mock Midterm = HW3 on Mon6; prepare for Midterm: Thu6.
Handed out: HW4 on Thu6; HW satisfaction quiz
Exam review sessions: TBD: Mon6, Tue6, and/or Wed6

Week 7: Arithmetic, Groups

Due: • HW4 on Fri7, 23:59
Handed out: DS3 on Wed7
RC: time and location TBD

Week 8: Graphs and Trees

Due:
- type your exercise from DS3 by Wed8,
- prepare to present during the Discussion session on Thu8,
- HW satisfaction quiz on Mon8.

Handed out: Mock Final = HW5 on Wed8
RC: time and location TBD

Week 9: Graphs and Trees, Review, Exams week with classes

Due: • HW5 on Tue9, 23:59
Handed out: nothing
RC: time and location TBD
Exam review sessions: TBD

This programme is long, and it is not set in stone. If you have a special interest in some of the topics and/or a lack of interest in others, and you can justify why it would be more useful for your future development to shift the focus or the order of priorities, don't hesitate to let me know! Discrete Mathematics covers a wide range of topics, and you can have a (limited) choice on what you learn in this course!

Main Course Textbook

Rosen, K. H., *Discrete Mathematics and its Applications*, 8th Ed., 2018, McGraw-Hill. ISBN 9781259676512.

Additional resources will be given at the appropriate time.

Recommended Literature

- Gallier, J., .*Discrete Mathematics*, Springer, New York, 2011, ISBN: 978-1-4419-8047-2
- Lovász, L., Pelikán, J., Vesztergombi, K. *Discrete Mathematics – Elementary and Beyond*, Springer, New York, 2003, ISBN: 978-0-387-21777-2
- Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, *Introduction to Algorithms*, 3rd ed., 2009, MIT Press, ISBN 978-0-262-03384-8 (hardcover : alk. paper)—ISBN 978-0-262-53305-8 (pbk. : alk. paper).

Acknowledgements: This course owes much of its structure and content to the textbook Rosen, K. H., *Discrete Mathematics and its Applications*, to the previous edition of the course at the UM-SJTU-JI taught by Prof. Runze Cai, to previous editions of the course taught at the University of Michigan, as well as other sources.

Outline

Ch. 0. Introduction

- Global Objectives
- Dramatis Personae – Meet a cast of intriguing problems!

Why study Discrete Mathematics?

Why study Discrete Mathematics?

- **Foundation for Computer Science and Engineering:**
  - Computers are fundamentally discrete.
  - Essential for algorithms, data structures, cryptography, networking, and software development.
- **Development of logical and analytical thinking**
  - Encourages problem-solving, critical thinking, and formal mathematical reasoning.
  - Gets you used to recognising structures of interest, patterns, structural invariants...

Applications of Discrete Mathematics

- **Real-World Use Cases**
  - **Cryptography**: Secure communication, encryption.
  - **Optimisation**: Routing, resource allocation, scheduling.
  - **Computer Networks**: Design and analysis of communication protocols.
- **Mathematical Rigor**
  - Focuses on proofs, formal definitions, and the structure of mathematical arguments.
- **Discrete Structures**
  - **Graphs**: Network analysis, pathfinding.
  - **Sets**: Operations in databases and data structures.
  - **Combinatorics**: Counting, arrangements, cryptography.

Long-Term Benefits

- **Prepares for Advanced Topics**
  - Theory of Computation, Graph Theory, Advanced Algorithms.
- **Cross-Disciplinary Connections**
  - Links to probability, statistics, philosophy, and more.
- **Real-World Problem Modelling**
  - Models systems in networks, simulations, and decision-making.

Dramatis Personae

Meet a cast of intriguing problems!

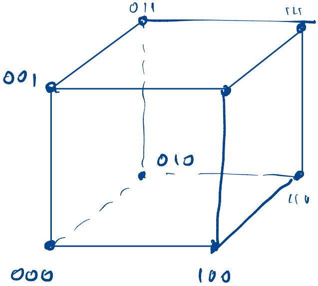
- A Game of Hats ___________?
- A Busy Postal Worker ___________?
- $i\mathcal{R}rational^{iRraTioNal} = \mathrm{rational?}$ ___________?
- An Infinite Hotel ___________?
- Smallest positive integer not definable in under sixty letters. ___________?
- The Blue-Eyed Islanders ___________?
- Colour everywhere! ___________?
- Chomp! ___________?
- How many musical notes can there be? ___________?

Objectives

- You are not expected to be able to solve the following problems easily.
- In a very low voice, discuss with the classmates adjacent to you how you might tackle such problems!
- What are we missing?
- What links may there be with the course contents?

A Game of Hats

- Ingredients: A mathematician: M; 3 students: A, B, C;
  3 white hats, and 3 black hats.
- During the next discussion session, M will place a hat on each of A, B, C, of colours $(h_1, h_2, h_3) \in \{W, B\}^3$.
- Each student will know the colour of the hats of the other two, but not their own.
- Students cannot communicate.
- At a given signal, each student can either:
  - Remain silent, or
  - Announce the colour they think their hat is.
- If no student speaks or any of the students is wrong, all receive -2 malus points.
- If at least one student speaks, and all students who speak are correct, they all receive $+1$ bonus point.



Can students expect to win at this game on average?

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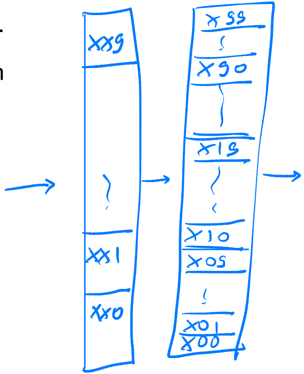
Would it help if they studied error-correcting codes?

A Busy Postal Worker

- You quit Discrete Math for a job at the Post Office.
- There's a huge heap of *n* letters in front of you, which you must sort by increasing postal code.
- How much space do you need to do it?
- How many comparisons will you need to perform?
- How many numbers do you wish to remember?



$i\mathcal{R}rationa\ell^{iRraTioNal} = \text{rational?}$

Are there irrational numbers x, y such that x^y is a rational number?

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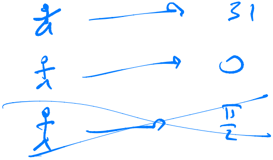
Are there irrational numbers x, y such that x^y is a rational number?
What if you don't need to actually find them?

- The Grand Countable, by Hilbert Hotels, has an infinite number of rooms: $0, 1, 2, \ldots$.
- Each room is numbered by a natural number, and each natural number has a room, for which it is the natural number that is the number of the room. [If you're Spanish, you know who speaks like this!]
- Anyway, back to our hotel: it's full.
- A finite number of guests arrives. Can we house them? If so, how? Can we do so while keeping only one guest per room?
- An infinite number of guests arrives. But you notice that each of them has a favourite natural number, different from the favourite number of all other guests! Can we still house them, one guest per room?
- What if their favourite numbers are rational instead of natural? What if they are real instead?



Smallest positive integer not definable in under sixty letters.

What is the "Smallest positive integer not definable in under sixty letters"?

- Which is it?
- Does it exist or not?
- Assume n is that number. Then what?

The Blue-Eyed Islanders

- An island has 1000 inhabitants, 100 of whom have blue eyes.
- Everyone knows the colour of the eyes of the others, but not their own.
- They never discuss eye colour, nor do they do anything that could result in them discovering the colour of their eyes.
- If an islander learns his eyes are blue, they must abandon the island the next day at noon.
- An outsider arrives on day 0, and remarks how surprised he is to find out that someone on the island has blue eyes, then leaves.

Does something change on the island?

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Does something change on the island?

What if there were only 1, 2, or 3 blue-eyed islanders?

Colour everywhere!

- You have coloured every single dot of the Euclidean plane either red, or green.
- (You were bored and found that pastime. We won't judge!)
- Prove that, however you may have coloured the plane, there is a rectangle of the plane whose 4 vertices are of the same colour.

Chomp!

- A deliciously deadly two-player game!
- The top left square c_{11} of a chocolate tablet is poisonous:

$$\begin{bmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ c_{21} & c_{22} & \dots & c_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ c_{m1} & c_{m2} & \dots & c_{mn} \end{bmatrix}.$$

- Taking turns with your frienemy, choose an un-eaten square of the chocolate tablet, and eat all squares to its right and below it. *E.g.*, if you pick square c_{ij}, you eat all c_{kl} for $i \leq k \leq m$ and $j \leq l \leq n$.

Can the first player win, no matter how the second plays? Is it the other way around? Or does it depend on n, m?

How many musical notes can there be?

- A taut metal string of length 1 vibrates when you strike or pluck it.
- It vibrates in a superposition of different modes, each of which produces different harmonics.
- Long story short, a string of length $1/2$ produces a note that we consider "the same" as the first, just an octave higher.
- A string of length $2/3$ produces a new note, at a higher pitch than the first (length 1). The interval those two notes form is called a "fifth". Together, they create a harmonious sound.
- You want to iterate that procedure: going up by another interval of a fifth to generate new notes that sound good with the previous ones.

How many notes do you generate this way?
Is 12 the only possible answer?
When should you stop?

Dramatis Personae

Meet a cast of intriguing problems!

- A Game of Hats — *Error-correcting code*
- A Busy Postal Worker — *Sorting algorithm*
- $i\mathcal{R}rational^{iRraTioNal} = \mathrm{rational}?$ — *Non-constructive proof*
- An Infinite Hotel — *Bijection, cardinality*
- Smallest positive integer not definable in under sixty letters. — *Logic*
- The Blue-Eyed Islanders — *Induction*
- Colour everywhere! — *The pigeonhole principle*
- Chomp! — *Dynamic programming*
- How many musical notes can there be? — *Groups, but not quite*